SREE CHAITHANYA INSTITUTE OF TECHNOLOGICAL SCIENCE

PROJECT SEMINAR

ON

IOT BASED COALMINE SAFETY MONITORING WITH HAZARD DETECTION AND REAL TIME ALERTS

Department of Electronics and Communication Engineering

PRESENTED BY

B.NANDINI(23TR1A0407)
G,RUCHITHAITHA(24TR5A0402)
S.PRANENDRA23TR1A0452)
R.GOURIPRIYA(23TR1A0447)
G,AKSHAYKUMAR(23TR1A420)
K.BHARATHKUMAR(23TR1A0431)
V.SUDHEER(23TR1A0461)

UNDER THE GUIDANCE OF
Mrs.NEETHIKA
ASSISTANT PROFESSOR

CONTENTS

INTRODUCTION

- This presentation explores the implementation of IoT-based systems designed To improve safety, IoT (Internet of Things) technology can be used to monitor coal mines in real-time, detect potential hazards, and alert authorities and miners.

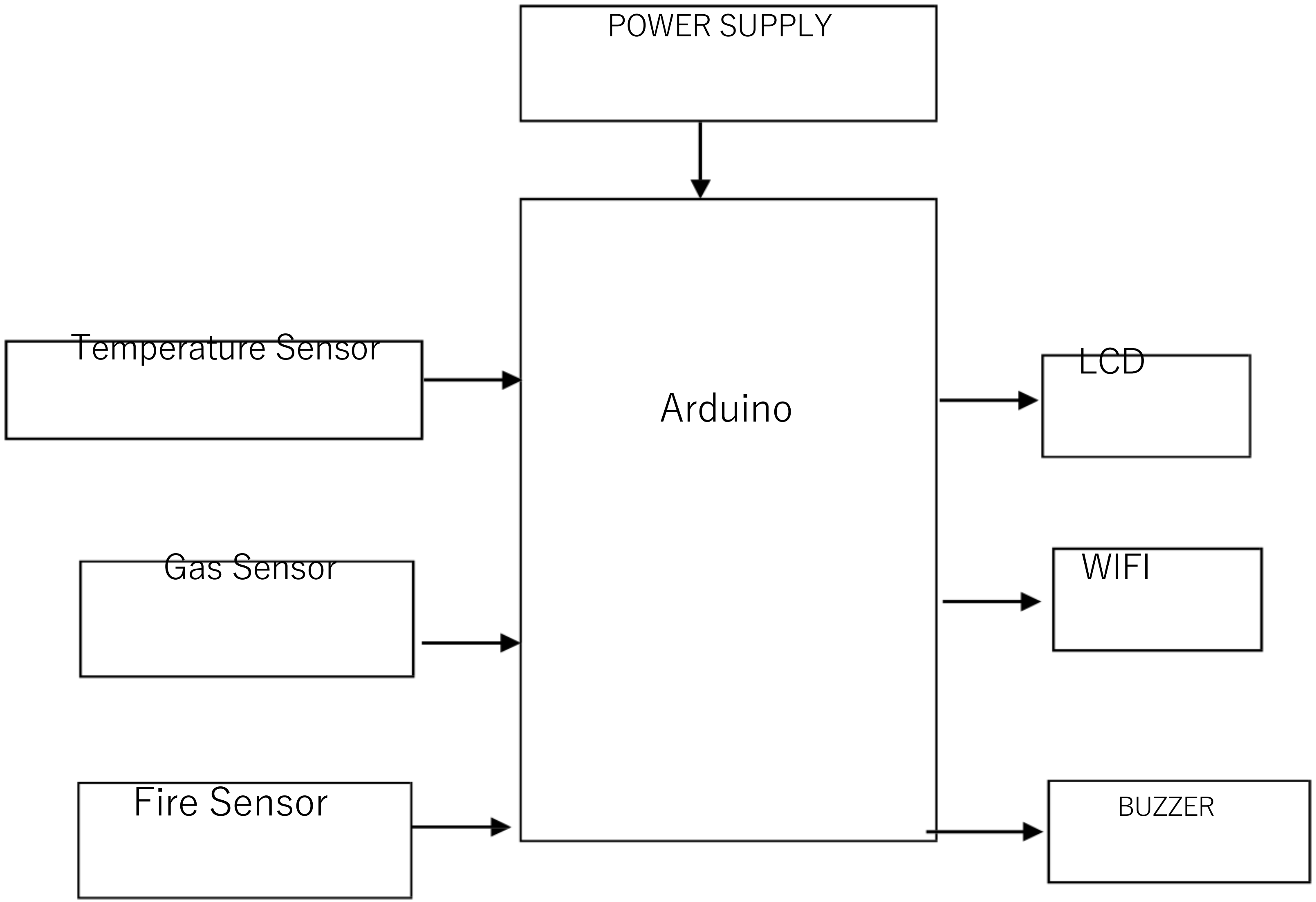
EXISTING SYSTEM

- Safety checks are done manually.
- No real-time alerts during danger.Only basic gas detectors are used.No system to watch data from far away.
- Alerts are not automatic (need human action).Not all hazards like fire or vibration are detected.
- Can be slow to respond in emergencies .Not suitable for modern safety need

PROPOSEDSYSTEM

- Uses sensors to detect gas, fire, temperature, and vibration .Sends real-time alerts through buzzer and mobile app.
- Shows data on a cloud or mobile dashboard.
- Works even without internet (buzzer gives alerts).Easy to install and low cost.
- Helps keep miners safe by warning early. Saves data for checking and analysis.
- Can be improved or expanded easily.

BLOCK DIAGRAM

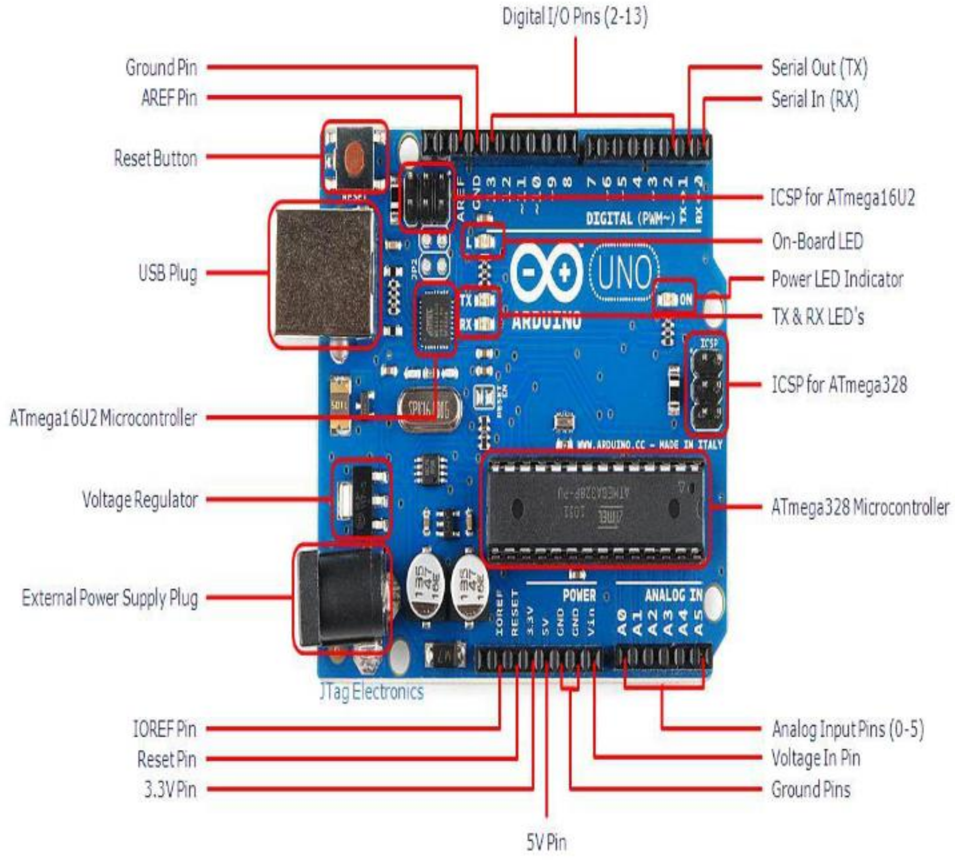


SYSTEM COMPONENTS

- ARDUINO UNO
- WIFI MODULE
- BUZZER
- LCD
- SENSORS
- RPS

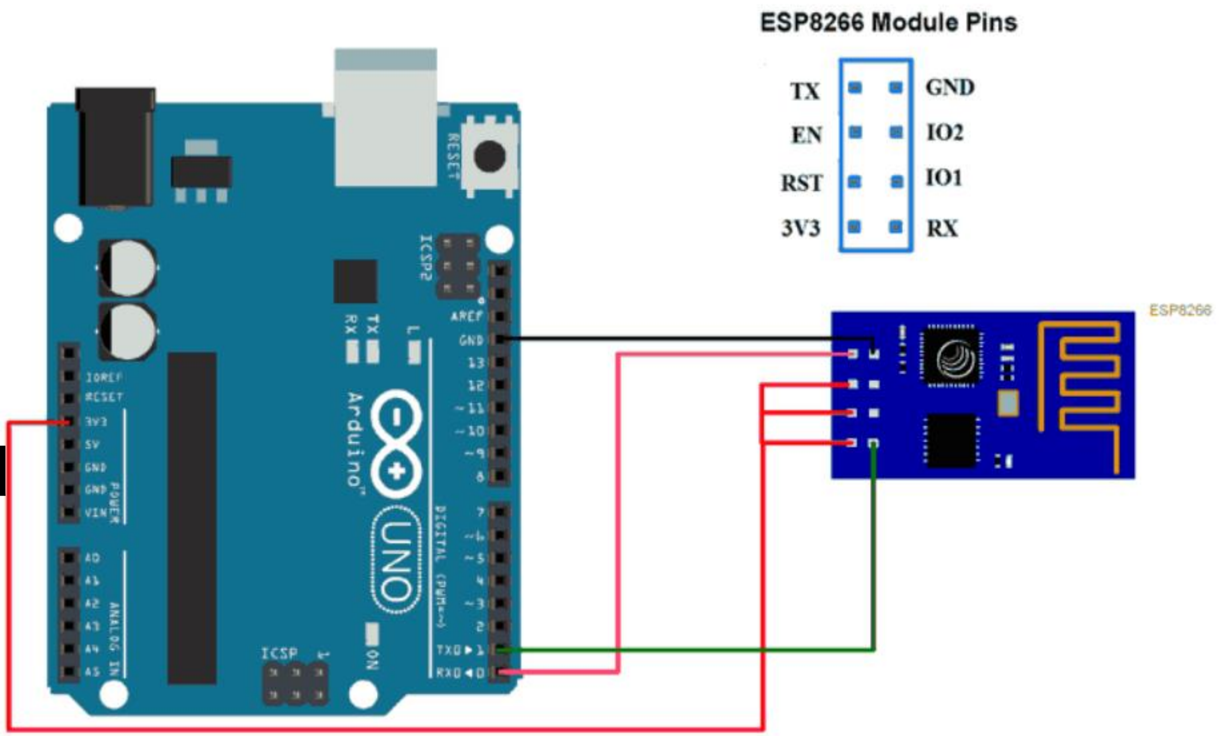
ARDUINO

- The Arduino Uno is a microcontroller board based on the ATmega328.
- It has 14 digital input/output pins
- 6 analog inputs
- 16 MHz ceramic resonator
- USB connection



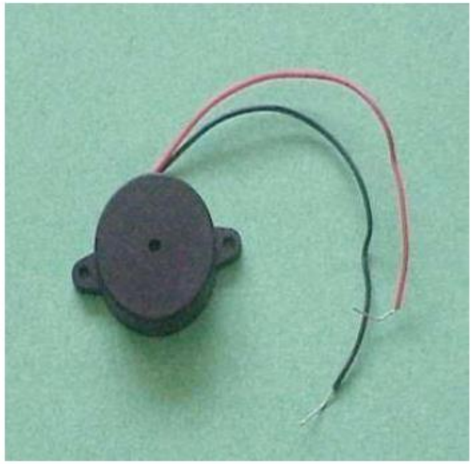
WIFI MODULE

- Connects to WiFi networks
- Transmits and receives data
- Enables remote monitoring and control.



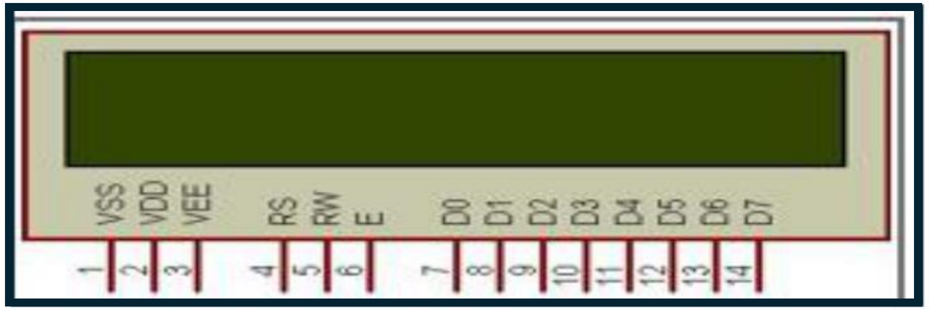
BUZZER

Buzzer is an Audio Signaling Device that Produces a Sound or Tone When an Electric Current is Applied

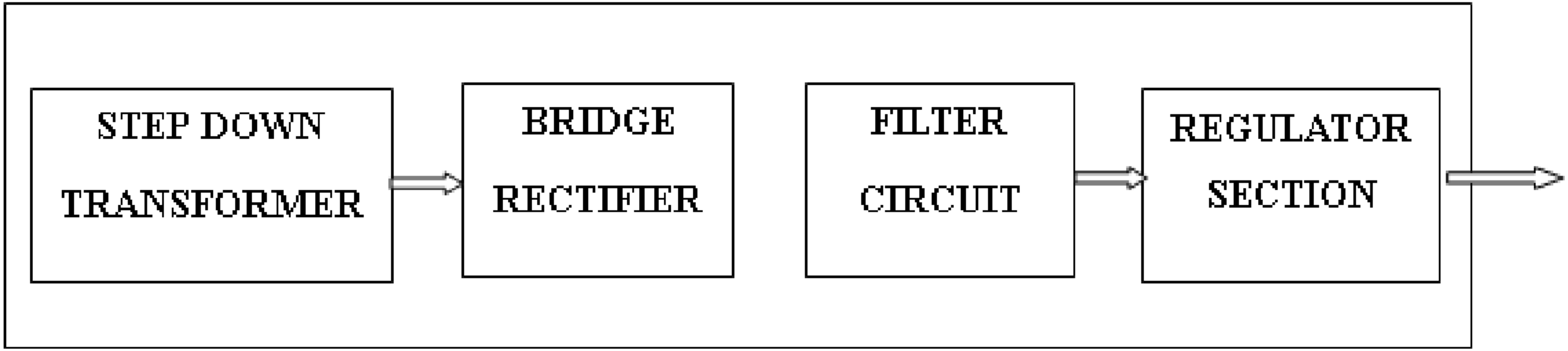


LCD

- Blocking or allowing light: Liquid crystals block or allow light to pass through a matrix of pixels
- Electric current: An electric current is applied to the liquid crystals to control their orientation
- Polarization: The liquid crystals change the polarization of light, which is then filtered to create images



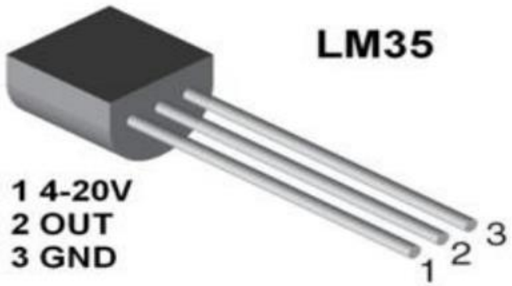
POWER SUPPLY SECTION



SENSORS

FLAME SENSOR

GAS SENSOR



TEMPERATUR
E

Working principle

- 1. Sensors detect hazards
- 2. Data is sent to a central system
- 3. System analyzes data
- 4. Alerts are sent if hazards are detected
- 5. Miners and supervisors take action to stay safe

APPLICATIONS

- Environmental Monitoring
- Gas Detection
- Temperature Monitoring
- Worker Safety Tracking
- Location-Tracking

ADVANTAGES

- Improved Safety
- Reduced accidents
- Increased

Productivity

- Real-time Alerts
- Data-driven Insights
- Cost Savings

FUTURE SCOPE

- . Intelligent Prediction:Integrate AI to analyze sensor data and predict hazards before they occur.
- Advanced Sensing:Add more sensors like dust, methane, and seismic detectors for deeper safety coverage.
- Wearable Technology:Develop smart wearables for miners to monitor health and send emergency alerts.
- Wider Applications:Adapt the system for use in other high-risk areas like tunnels, oil rigs, and chemical plants

CONCLUSION

- The system helps detect hazards like gas, fire, and heat in real-time.
- It gives quick alerts to protect miners and reduce accidents.
- The project is low-cost, reliable, and useful for improving safety.

THANK YOU